

Figure 4.2: Architecture Work Flow (Different Layers)

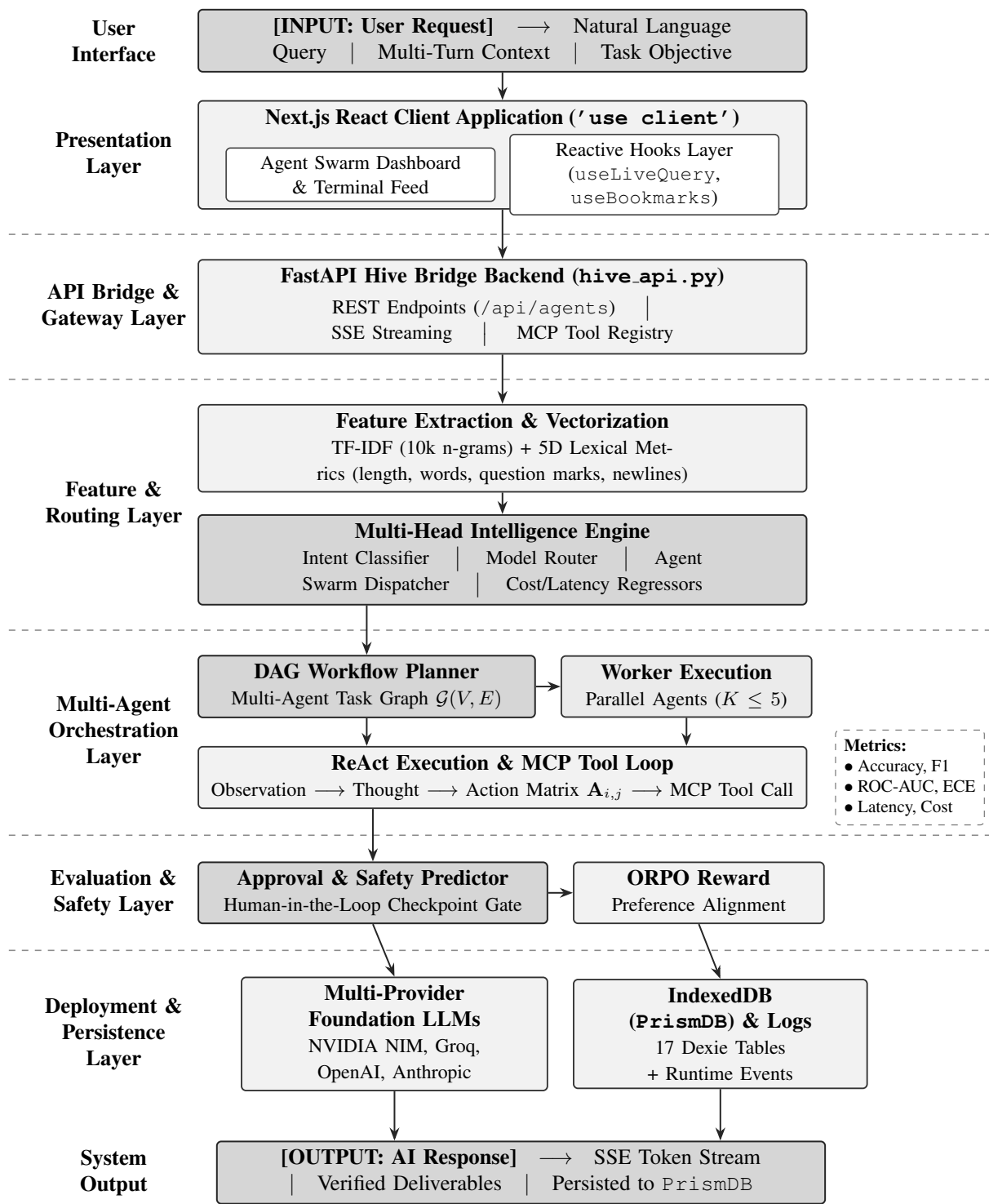


Figure 4.2: Different Layers.

Glossary of Terms & Abbreviations:

- **FastAPI:** High-performance asynchronous Python web framework for backend APIs.
- **REST:** Representational State Transfer (standard HTTP API design pattern).
- **SSE:** Server-Sent Events (persistent HTTP connection for real-time text streaming).
- **MCP:** Model Context Protocol (open standard for agent tool and resource discovery).
- **DAG:** Directed Acyclic Graph (execution plan with task dependency order, no cycles).

- **ReAct:** Reason + Act (agent loop alternating between reasoning and tool calls).
- **ORPO:** Odds Ratio Preference Optimization (direct model alignment without reward model).
- **IndexedDB:** Transactional browser database for client-side persistence (PrismDB).
- **Dexie.js / useLiveQuery:** Reactive JavaScript engine updating UI on DB changes.
- **ROC-AUC / ECE:** Classification discrimination metric / Expected Calibration Error.